1 Unified Valuation and Global Objective Framework

This chapter presents the core valuation and optimization framework that integrates competitive performance, financial constraints, and long-term brand equity into a single global objective function Z. By unifying short-term cash flow (Π_t) and long-term franchise value growth ($\Delta V_{\text{franchise},t}$), the model enables dynamic decision-making that balances "wins" and "wealth" in a large-market context such as the Los Angeles Lakers.

1.1 Multi-Criteria Design of the Global Objective Function Z

The overarching goal is to maximize the expected discounted total value over a planning horizon T (typically 3–5 years), accounting for time preference and the trade-off between immediate profitability and sustained franchise appreciation.

The global objective function is formulated as:

$$Z = \max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t \left(w_1 \cdot \Pi_t + w_2 \cdot \Delta V_{\text{franchise},t}\right)\right] \tag{1}$$

where:

- $\gamma = 0.92$ is the annual discount factor. This reflects an approximately 8% time preference rate, calibrated to large-market owner patience [3], and is consistent with sports franchise Discounted Cash Flow (DCF) models.
- $w_1 = 0.45$ and $w_2 = 0.55$ are the weights on short-term operational profit Π_t and long-term franchise value increment $\Delta V_{\text{franchise},t}$, respectively.
- $m = 18.15$ is the revenue multiple for large-market teams like the Lakers (calculated as \$10 billion valuation / \$551 million revenue in 2025 [2]).

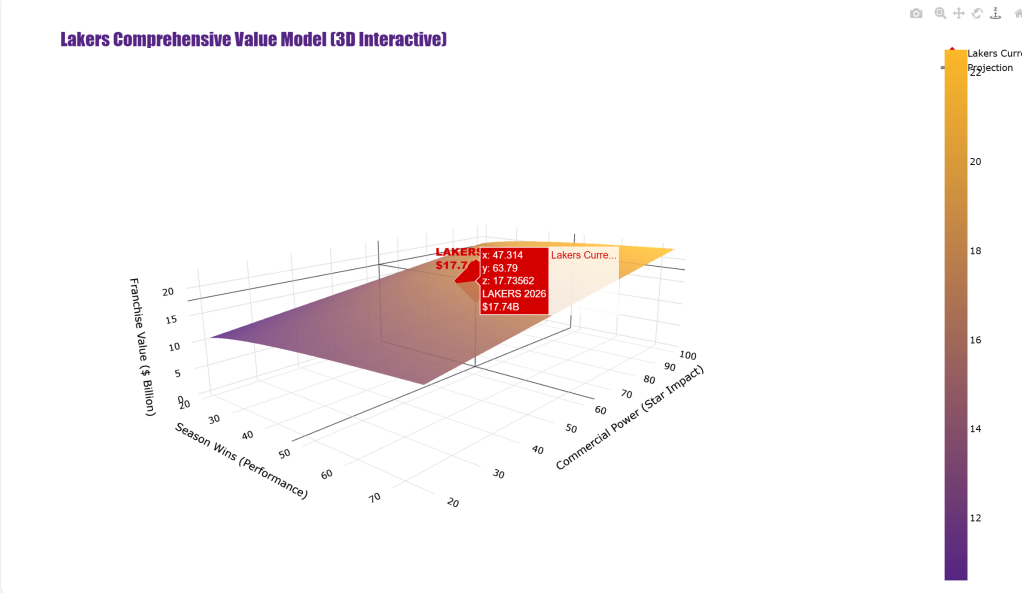


Figure 1: 3D Visualization of the Comprehensive Value Model, illustrating the interaction between competitive success, star impact, and total franchise valuation.

Table 1: Key Parameters in Global Objective and Player Valuation

Parameter	Value	Role/Interpretation	Calibration Source
γ	0.92	Annual discount factor	Large-market DCF models
w_1/w_2	0.45 / 0.55	Short-term vs long-term value weights	Lakers revenue attribution
$w_{\text{on}}/w_{\text{off}}$	0.45 / 0.55	On-court vs off-court player value weights	Large-market premium
m	18.15	Revenue multiple	Forbes 2025 ($10B / $551M)

1.2 Deconstruction of Total Player Value (V_{total})

Player valuation serves as the foundational micro-level input to the framework. Each player's total value V_{total} is decomposed into on-court competitive contribution (V_{on}) and off-court commercial impact (V_{off}), standardized to a 0–10 scale for aggregation and comparison.

The total player value is defined as:

$$V_{\text{total}} = w_{\text{on}} \cdot V_{\text{on}} + w_{\text{off}} \cdot V_{\text{off}} \tag{2}$$

where $w_{\text{on}} = 0.45$ represents the baseline competitive contribution, and $w_{\text{off}} = 0.55$ represents the commercial value premium. This weighting is calibrated to reflect the Lakers' heavy reliance on ticket sales, merchandise, sponsorships, and social media-driven revenue [3].

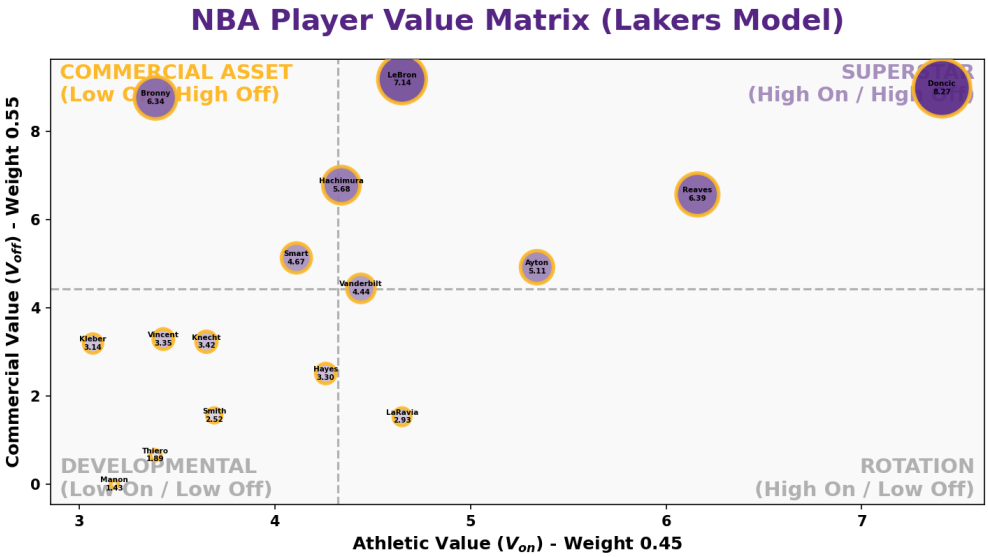


Figure 2: NBA Player Value Matrix: Categorizing players based on Athletic (V_{on}) and Commercial (V_{off}) dimensions.

To visualize this hierarchical structure and weight distribution, the decomposition of player value components is illustrated in Figure 3.

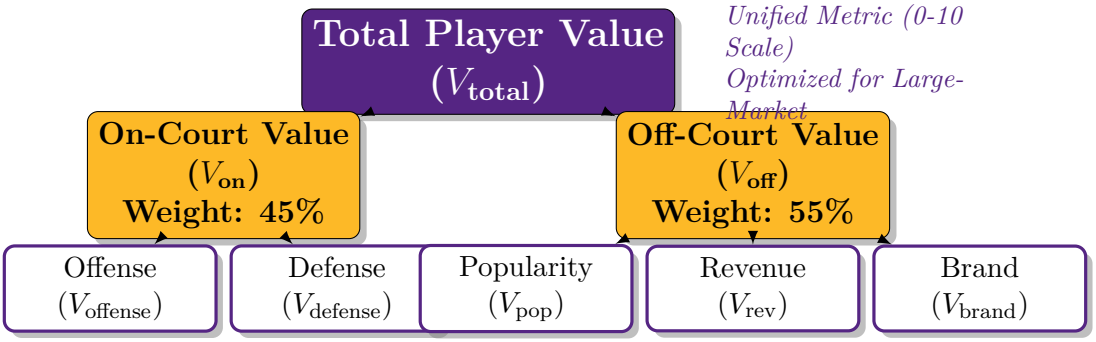


Figure 3: Decomposition of Total Player Value (V_{total}), highlighting the heavy weighting on Off-Court Commercial Value (55%) characteristic of large-market franchises (Lakers Theme).

1.2.1 Off-Court Commercial Value (V_{off}) and Star Impact

V_{off} quantifies non-game revenue contribution through three sub-modules. Crucially, high V_{off} scores drive the "Star Count" metric used in brand valuation:

Definition of Star Impact (N_{stars}): N_{stars} is defined as the count of active players with $V_{\text{off}} \geq 8.0$ (high commercial impact threshold), reflecting superstar-driven revenue dominance.

The specific sub-modules for V_{off} are calculated as follows:

$$V_{\text{off}} = w_1 \cdot V_{\text{pop}} + w_2 \cdot V_{\text{rev}} + w_3 \cdot V_{\text{brand}} \tag{3}$$

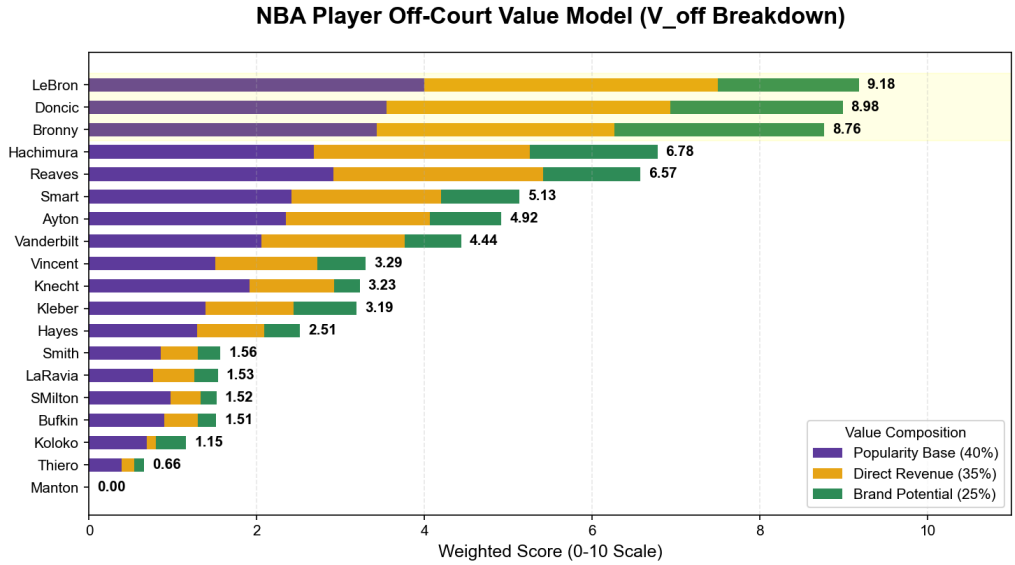


Figure 4: Breakdown of Off-Court Commercial Value (V_{off}) for key players, illustrating the contribution of popularity and revenue streams.

Popularity Base (V_{pop}):

$$V_{\text{pop}} = 10 \cdot \text{Social}_{\text{std}} + 12 \cdot \text{Jersey}_{\text{std}} + 8 \cdot \text{Media}_{\text{std}} \tag{4}$$

where *Social* aggregates followers across major platforms (Instagram, X, TikTok), *Jersey* is the inverse NBA official sales ranking (Rank 1 = 10), and *Media* tracks exposure volume.

Direct Revenue Contribution (V_{rev}):

$$V_{\text{rev}} = 10 \cdot \text{Attend}_{\text{std}} + 9 \cdot \text{Sponsor}_{\text{std}} + 7 \cdot \text{Ticket}_{\text{std}} \tag{5}$$

where *Attend* measures the road attendance boost percentage, *Sponsor* captures individual brand-attributed premiums, and *Ticket* reflects contribution to secondary market pricing.

1.2.2 On-Court Competitive Value (V_{on})

V_{on} captures on-court impact through offensive and defensive sub-components. Before aggregation, these components are defined as follows:

Aggregated On-Court Value The final V_{on} score is strictly adjusted for durability risks:

$$V_{\text{on}} = (w_{\text{offense}} \cdot V_{\text{offense}} + w_{\text{defense}} \cdot V_{\text{defense}}) \times \Phi \times (1 - P_{\text{risk}}) \tag{6}$$

where $w_{\text{offense}} = 0.50$, $w_{\text{defense}} = 0.50$ (balanced configuration), and Φ is the durability factor $(1 - \text{injured_games/projected_games})$.

Offensive Value (V_{offense})

$$V_{\text{offense}} = 0.35V_{\text{core}} + 0.30V_{\text{eff}} + 0.25V_{\text{usage}} + 0.10V_{\text{trans}} \tag{7}$$

- **Core:** OWS (Offensive Win Shares) + OBPM (Offensive Box Plus/Minus).
- **Efficiency:** TS% (True Shooting Percentage) + PER+ (adjusted for turnovers: $1 - \text{TOV\%}$).
- **Volume:** USG% (Usage Rate) + 3PM + AST% (Correlated with commercial appeal).

Defensive Value (V_{defense})

$$V_{\text{defense}} = 0.35V_{\text{core_def}} + 0.30V_{\text{output}} + 0.25V_{\text{indiv}} + 0.10V_{\text{rot}} \tag{8}$$

- **Core:** DWS (Defensive Win Shares) + DBPM.
- **Output:** STL (Steals) + BLK (Blocks) + DRB% (Defensive Rebound Rate).
- **Individual:** Diff% (Opponent FG% differential) + Defensive Impact.

Risk Model and Injury Probability (P_{risk}) Instead of a simple linear penalty, we employ a Logistic Regression model to predict the probability of significant absence in the next season. The probability is calculated as:

$$P_{risk} = \frac{1}{1 + e^{-Z}} \tag{9}$$

Where the Log-Odds (Z) is defined by the following predictors:

$$Z = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{History}) + \beta_3(\text{USG\%}) + \beta_4(\text{MPG}) + \beta_5(\text{Pos}) \tag{10}$$

Table 2: Logistic Regression Coefficients for Injury Prediction

Parameter	Coeff (β)	Predictor Logic & Scope
β_0 (Intercept)	-5.0	Baseline. Low base probability for replacement level.
β_1 (Age)	0.15	**Age Factor:** Positive correlation. Odds increase $\approx 15\%$ per year.
β_2 (History)	0.80	**Prior Injuries:** Count of major injuries per season.
β_3 (USG%)	0.05	**Usage Rate:** Workload proxy. High usage increases mechanical stress.
β_4 (MPG)	0.03	**Minutes/Game:** Endurance load. Positive correlation with high load.
β_5 (Pos)	+0.6 / -0.5	**Position Dummy:** Center/PF = +0.6; Guard = -0.5.

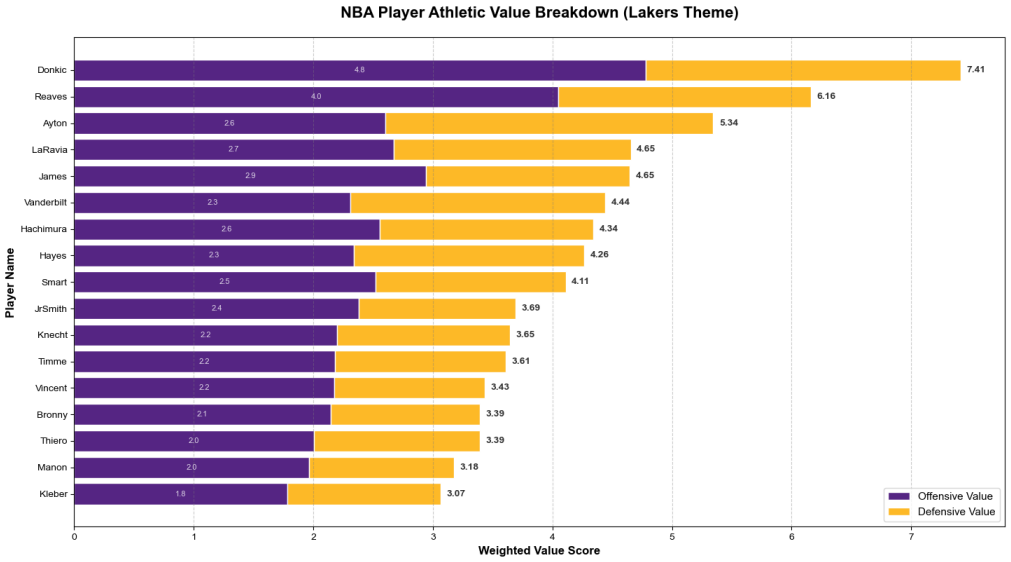


Figure 5: Distribution of Athletic Value (V_{on}), showing the balance between offensive and defensive contributions.

1.3 Standardization Note

All raw data (X_{raw}) must be mapped to a 0-10 scale, using league-wide statistics as the baseline:

$$X_{std} = 5 + 5 \times \frac{X_{raw} - \mu_{league}}{\sigma_{league}} \tag{11}$$

Note: For negative indicators such as Injury Probability (P_{risk}) or Turnover Rate, a reverse mapping strategy is applied.